Spandana Madi Reddy

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SUMMARY

Experienced Machine Learning Engineer with over 3+ years of experience architecting and deploying predictive models across insurance and risk domains, leveraging tools like XGBoost, TensorFlow, and Amazon SageMaker to solve real-world business problems. Adept at leading cross-functional collaboration, automating end-to-end ML pipelines using Docker and Apache Airflow, and delivering scalable solutions through advanced data engineering, model interpretability, and real-time API integration.

SKILLS

Programming Languages: Python (NumPy, Pandas, Regex, Pickle, OS, JSON), SQL (PostgreSQL, MySQL)
Machine Learning Algorithms: Linear Regression, Logistic Regression (Binary & Multiclass), Ridge/Lasso/Polynomial Regression, Decision Tree, Random Forest, SVM, KNN, Naive Bayes, GBM, XGBoost, LightGBM, CatBoost, Statsmodels
ML Libraries & Frameworks: Scikit-learn, TensorFlow, spaCy, NLTK, PyCaret, Matplotlib, Seaborn
Model Deployment & MLOps: Amazon SageMaker, Docker, Apache Airflow, Flask, FastAPI, AWS Lambda
Data Handling & Analysis: ETL, Data Cleaning, Feature Engineering, Exploratory Data Analysis (EDA), Structured & Unstructured Data, Joins, Aggregations
Data Visualization: Tableau, Power BI, Advanced Excel, Looker
Development Environments: Jupyter Notebooks, PyCharm, Microsoft Visual Studio
Collaboration & Reporting: JIRA, Confluence, Microsoft Teams, TestRail

EXPERIENCE

Liberty Mutual Insurance, USA | Machine Learning Engineer **Jan 2025 – Present**

- Engineered and deployed an ML pipeline integrating image and text data from historical claims to predict vehicle damage severity and payout ranges, leveraging TensorFlow, XGBoost, and NLP with Scikit-learn and OpenCV for multimodal data fusion.
- Designed and operationalized a Convolutional Neural Network to classify accident images and ensemble models for payout estimation, improving claim consistency across adjusters and enhancing prediction reliability by over 30%.
- Spearheaded real-time model serving via Docker and AWS SageMaker while automating workflows using Apache Airflow, resulting in 40% faster claim resolution and seamless integration into the Liberty Auto Claims Management Suite.
- Collaborated with cross-functional teams to analyze model performance through Tableau dashboards and feedback loops, which enabled identification of model drift and led to a retraining pipeline that improved payout accuracy by 25%.
- Led the development of a Smart Claim Estimator used by Liberty's mobile app, enabling users to receive instant damage evaluations; this feature increased digital claim pre-approvals by 35% and reduced manual interventions by 40%.

AIG, USA | Machine Learning Engineer **May 2024 – Dec 2024**

- Designed and deployed a scalable ML pipeline using XGBoost, PostgreSQL, and Apache Airflow to automate commercial risk scoring, classifying applicants into high, medium, and low risk, while integrating structured and unstructured data from internal and external sources.
- Implemented advanced NLP techniques using spaCy and NLTK to extract risk-sensitive keywords from safety reports and application narratives, enabling contextual understanding and enhancing feature depth for the classification model.
- Containerized and served the model on Amazon SageMaker via REST API, facilitating real-time integration with the underwriting workflow and improving risk evaluation consistency across departments by over 35%.
- Collaborated with data and underwriting teams to develop a Power BI dashboard and integrate feature importance outputs, which reduced underwriting time by 45% while improving high-risk applicant precision by 87%.
- Increased model adoption by creating a weekly retraining pipeline using Apache Airflow and Docker that boosted model accuracy by 12% and improved feature-based transparency across the RiskGuard platform by 40%.

Experion Technologies, India, Machine Learning Engineer **Aug 2021 – Jul 2023**

- Developed and deployed predictive models using XGBoost and LightGBM to classify policyholders and claims into risk categories, integrating PostgreSQL and MongoDB for structured and unstructured data ingestion.
- Engineered fraud detection logic by combining Isolation Forest and supervised binary classifiers with real-time Kafka streaming, resulting in automated claim scoring pipelines within client claim portals.
- Designed and exposed model endpoints using Flask APIs and Docker containers, integrating with CRM and fraud review tools, while orchestrating retraining workflows via Apache Airflow for consistent performance.
- Leveraged SHAP and Power BI dashboards to interpret model outputs, enabling insurance agents and fraud reviewers to make data-driven decisions and prioritize high-risk cases with greater transparency.
- Conducted advanced feature engineering by analyzing app usage, communication patterns, diagnosis irregularities, and policy-payment behaviors to enrich training data and improve model accuracy.
- Improved policyholder retention rate by 26% and reduced fraudulent claim payouts by 35% by building ML solutions that proactively identified lapse-prone users and anomalous claim submissions during live processing.

EDUCATION

Masters in Information Technology, University of Cincinnati, Cincinnati, Ohio **Dec 2024**
Bachelors in Information Technology, Sri Indu College of Engineering and Technology, India **April 2023**